

Ayan Kumar

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SUMMARY

Full Stack Developer and AI/ML enthusiast building production-grade web apps, CLI tooling, and LLM-powered systems. Proficient in TypeScript, Next.js, Node.js, and Python with hands-on experience in RAG pipelines, vector databases, and multi-provider LLM integrations. Strong DSA fundamentals and open source contributor.

EDUCATION

Punjabi University, Patiala
B.Tech in Computer Engineering

Patiala, Punjab
Jul 2024 – Jul 2028

EXPERIENCE

- Open Source Contributor – GSSoC 2026** May 2026 – Present
Rocket.Chat, Python Ecosystem, Software Mansion Remote
– Contributing bug fixes, feature additions, and documentation improvements across multiple repositories on Github.
- AI/ML Intern** Dec 2025 – Jan 2026
ShadowFox Remote
– Built end-to-end ML pipelines using TensorFlow with hyperparameter tuning; preprocessed 100K+ records using Pandas/NumPy for downstream model training.
– Collaborated on A/B testing setup and real-time monitoring dashboards for production ML deployments in Agile environment.

PROJECTS

- Smart Local Knowledge Assistant** | *TypeScript, Next.js 16, Gemini API, Vectra, pdf-parse*
– Built RAG-based AI assistant grounding responses in verified Indian government documents via local Vectra vector DB – eliminates hallucinations with source-cited citation cards.
– Implemented on-the-fly ingestion pipeline: .pdf/.txt uploads are chunked, embedded, and indexed in real time with regional filtering by State and District.
– Integrated Gemini Pro/Flash with dynamic model routing and fallback handling; achieved sub-2s response time. Live: smart-assistant-taupe.vercel.app
- LLMVC – LLM Prompt Version Control CLI** | *TypeScript, Node.js, SQLite, Ollama, OpenRouter*
– Local-first CLI providing git-style prompt versioning (save, diff, rollback) via SQLite across 5+ LLM providers (OpenRouter, Ollama, Gemini).
– Word-level diff engine + AI-powered prompt scoring (clarity, specificity, context) using LLM-as-judge eval runner with Qwen3 via Ollama.
– Debugged critical multi-provider issues: wrong API auth headers, incorrect endpoint routing, and Gemini socket hang-up errors – restoring full provider support. github.com/kumar-ayan/LLMVC
- Anonymous Location-Based Social Platform** | *React Native, Firebase/Supabase, Mapbox API*
– Architected privacy-focused platform for anonymous geotagged posts – manual location selection, no exact user tracking; designed real-time data handling and anonymous auth systems.
– Designed gamification mechanics (achievement badges, exploration rewards, engagement-based progression) to drive user retention and platform engagement.
– Planned scalable backend architecture with privacy-preserving analytics and premium monetization features including enhanced customization and unlimited media uploads.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks: React.js, Next.js, Node.js, React Native, Tailwind CSS, Django
AI/ML: Gemini API, TensorFlow, scikit-learn, HuggingFace, RAG, Vector Embeddings, Ollama, OpenRouter, LangChain
Databases & Tools: SQLite, MongoDB, Git, Vercel, Postman, Jupyter

ACHIEVEMENTS

Solved 300+ DSA problems on LeetCode/GeeksforGeeks – dynamic programming, graphs, trees, greedy algorithms.
Contributor to **Gitree** – PyPI-published Python CLI tool for LLM context generation from codebases.
Certifications: Elements of AI, GitHub Copilot Integration, EA Software Engineering Virtual Experience, Gen AI Academy 2.0.